

Test Strategy Document

Project Information

Item	Details
Project Name	AI/ML Web Application
Document Version	1.0
Prepared By	QA Tester Intern
Date	18 June 2026
Testing Type	Functional, Automation, Performance, API, Regression
Objective	Ensure application quality, reliability, performance, and security before release

1. Purpose

The purpose of this Test Strategy is to define the testing approach, scope, objectives, resources, tools, and deliverables for validating the AI/ML application.

The strategy ensures that all critical functionalities are tested thoroughly and meet business requirements.

2. Testing Objectives

Primary Objectives

- Verify all application functionalities work correctly.
- Validate user workflows.
- Ensure APIs return expected responses.
- Confirm system stability under load.
- Identify and document defects.
- Reduce production risks.
- Improve overall software quality.

3. Scope of Testing

In Scope

Authentication Module

- User Registration
- Login
- Logout
- Password Reset
- Session Management

Dashboard Module

- Navigation
- Widgets
- Reports
- User Preferences

AI/ML Features

- Data Upload
- Data Processing
- Prediction Generation
- Model Execution
- Result Visualization

API Services

- GET Requests
- POST Requests
- PUT Requests
- DELETE Requests
- Error Handling

Performance

- Response Time
- Concurrent Users
- Resource Usage

Out of Scope

- Third-party payment gateways
 - Infrastructure testing
 - Source code review
 - Hardware testing
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4. Testing Types

Functional Testing

Purpose:

Verify business requirements are met.

Coverage:

- UI Validation
- Input Validation
- Workflow Validation
- Data Validation

Expected Outcome:

All functionalities work as intended.

Regression Testing

Purpose:

Ensure new changes do not affect existing features.

Coverage:

- Login
- Dashboard
- Reports
- APIs

Expected Outcome:

Existing functionality remains stable.

API Testing

Purpose:

Validate backend communication.

Coverage:

- Status Codes
- Request Validation
- Response Validation
- Error Messages

Tools:

- Postman
- ReqRes API

Expected Outcome:

Correct responses and status codes.

Automation Testing

Purpose:

Reduce repetitive manual effort.

Coverage:

- Login
- Registration
- Navigation
- Form Submission

Tools:

- Selenium
- Pytest

Expected Outcome:

Automated scripts execute successfully.

Performance Testing

Purpose:

Measure application behavior under load.

Coverage:

- Load Testing
- Stress Testing
- Response Time Analysis

Tools:

- JMeter

Expected Outcome:

Application handles expected user traffic.

5. Test Environment

Component	Configuration
OS	Windows 10/11
Browser	Chrome
API Tool	Postman
Automation Tool	Selenium
Performance Tool	JMeter
Version Control	GitHub

6. Entry Criteria

Testing begins when:

- Requirements are available
- Test environment is ready
- Build is deployed
- Test data is prepared

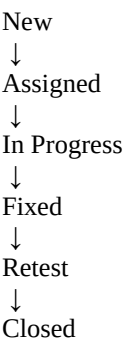
7. Exit Criteria

Testing is completed when:

- All critical test cases pass
- No Critical Severity defects remain open
- Regression testing passes
- Test Summary Report is prepared
- Stakeholder approval received

8. Defect Management Process

Defect Lifecycle



If issue persists:



9. Severity Levels

SEVERITY	DESCRIPTION
CRITICAL	System crash/data loss
HIGH	Major functionality broken
MEDIUM	Feature partially affected
LOW	UI/Cosmetic issue

10. Risk Analysis

Risk	Impact	Mitigation
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Requirement Changes	High	Frequent reviews
Environment Issues	Medium	Early setup
Delayed Build	High	Daily coordination
Limited Test Data	Medium	Prepare sample datasets

11. Deliverables

The QA team will provide:

- Test Strategy Document
 - Test Plan
 - Test Cases
 - Automation Scripts
 - API Test Reports
 - Defect Reports
 - Regression Test Reports
 - Performance Test Reports
 - Final Test Summary Report
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12. Test Metrics

Metric	Target
Test Case Pass Rate	>95%
Defect Leakage	<5%
Critical Defects	0
Response Time	<3 sec
Automation Coverage	>70%

13. Resource Allocation

Role	Responsibility
QA Tester Intern	Test Design, Execution, Reporting
Developer	Bug Fixing
Project Lead	Review & Approval

14. Final Recommendation

The application will be considered **Ready for Release** when:

- All critical functionalities pass testing.

- No Critical or High severity defects remain open.
- Performance benchmarks are achieved.
- API validations are successful.
- Regression testing confirms system stability.

Prepared By: QA Tester Intern

Version: 1.0

Status: Ready for Review